$Dac := 2.5$ $\quad$ $Totmile := .775$

$Dbc := 4.5$ $\quad$ $j := \sqrt{-1}$

$Dca := 7.0$

$Deq := \sqrt{Dac \cdot Dbc \cdot Dca}$ $\quad$ $pf := .9$

$$Zline := .592 + j \cdot .12134 \cdot \ln\left(\frac{Deq}{.00814}\right) \cdot Totmile$$

$Zline = 0.592 + 0.658i$ $\quad$ $arccos := \cos^{-1}$

$V := 4160$

$ToTpower := 1675$

$$I := \frac{1675}{\sqrt{3} \cdot 4160} \cdot arccos(pf)$$

$Vdrop := Zline \cdot I$

$Vdrop = 0.062 + 0.069i$

$$Vln := \frac{4160}{\sqrt{3}}$$

$$Kdrop := \frac{.062}{Vln} \cdot 100$$ $\quad$ $$Krise := \frac{.069}{Vln}$$

$Kdrop = 0.003$ $\quad$ $Krise = 2.873 \cdot 10^{-5}$

$Vdrop1 := Kdrop \cdot ToTpower \cdot .15$ $\quad$ $Vdrop1 = 0.649$

$Vdrop2 := Kdrop \cdot 1475 \cdot .175$ $\quad$ $Vdrop2 = 0.666$

$Vdrop3 := Kdrop \cdot 1325 \cdot .2$ $\quad$ $Vdrop3 = 0.684$

$Vdrop4 := Kdrop \cdot 1225.125$ $\quad$ $Vdrop4 = 3.163$

$Vdrop5 := Kdrop \cdot 925 \cdot .225$ $\quad$ $Vdrop5 = 0.537$

$Vdrop6 := Kdrop \cdot 500 \cdot .125$ $\quad$ $Vdrop6 = 0.161$

$Vdr4 := Kdrop \cdot 500 \cdot .35$

$Vrs4 := Vdr4 \cdot .03$

$$KVARc := \frac{Vrs4}{Krise}$$

$KVARc = 471.739$

Non-Commercial Use Only